Proceedings of the 4th Conference on Fully Actuated System Theory and Applications
July 4-6, 2025, Nanjing, China

# Sliding Mode Control for Flexible Joint Space Robot Via Nonlinear Integration

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**Abstract:** Aiming at solving the problems of trajectory tracking and joint flexible vibration of base attitude-controlled flexible jointed space robots under external disturbances, a novel control method based on nonlinear integral sliding mode is proposed. Firstly, the dynamic model of flexible joint space robot with floating base attitude controlled is established by using Lagrange method, and the system model is approximately decomposed into two subsystems, which are fast and slow, according to singular perturbation theory. For the slow subsystem, a nonlinear integral sliding mode surface and a fast reaching law are introduced to design the sliding mode controller, which improves the robustness of the system and weakens the chattering phenomenon. A speed difference feedback control algorithm is designed for the fast subsystem to suppress the vibration of the flexible joints and ensure the control accuracy. Finally, the effectiveness of the method is verified by simulation.

**Key Words:** Robustness, Singular perturbation method, Integral sliding mode, Flexible joint space robot

2025 4th Conference on Fully Actuated System Theory and Applications (FASTA) | 979-8-3315-2692-4/25/$31.00 ©2025 IEEE | DOI: 10.1109/FASTA65681.2025.11138780

## 1 Introduction

With the deepening of human space exploration, space robots have played an important role in maintaining and recovering satellites, maintaining and repairing spacecraft, constructing and assembling space stations and performing dangerous tasks in place of astronauts, and have attracted widespread attention [1-3]. With the enhancement of space exploration technology, the requirements for space robots are increasing, and the scope of application about flexible joint space robots is becoming more and more widespread.

The flexibility of the joints can make the space robot absorb the collision energy in order to avoid damaging the actuator when capturing the target load, but its flexible characteristic will cause residual vibration at the end of the manipulator, which will easily lead to the motor angle at the joints and the arm angle can not be synchronized, and this will generate the angular error, which seriously affects the accuracy of the controller, and influences the stability of the system and the control accuracy [4,5]. Therefore, the vibration suppression and high-precision control of flexible joint space robots are becoming a hot research topic for scholars at home and abroad [6].

Many scholars have designed different high-performance controllers to improve the control performance of flexible joint space robot. In [7], the article introduced given performance constraints into robust control law for free-flying flexible joint space robot by using Backstepping technique and optimized with dynamic surface control technique, and a disturbance observer was designed for estimating the unknown time-varying perturbations. The singular perturbation theory can decompose the system into "slow dynamics" and "fast dynamics" two subsystems, to analyze and design respectively, on the one hand, controlling the trajectory of the system to track the desired trajectory, and on the other hand, suppressing the elastic vibration of the joints [8,9]. In [10], by applying singular perturbation method, the system was divided into fast and slow subsystems, the torque differential feedback control algorithm was used for the fast subsystem and optimization algorithm for adaptive control with radial basis function neural network was applied in slow subsystem. In [11], a robust controller based on sliding observer was proposed for the slow subsystem, otherwise, optimal linear quadratic regulator method was used to design the controller by using the linear observer to estimate the speed in fast subsystem.

Sliding mode variable structure control has attracted much attention from scholars due to its strong robustness [12]. An improved sliding mode control method was proposed in [13], for slow subsystem, the computed torque method was designed to optimize the sliding mode control algorithm and ensure the real-time performance of the system, a PID controller was designed to achieve the rigidity compensation and suppress the elastic vibration for fast subsystem. In [14], a sliding mode variable structure control that was improved by the least squares method for quality identification was suggested, in order to ensure the control accuracy, a hyperbolic tangent sliding mode variable structure controller was designed for slow subsystem, and a fast subsystem feedback compensation control algorithm based on speed difference is designed, which serves to restrain the residual vibration created by flexible joints. However, the sliding mode control produces chattering due to the high frequency switching of the discontinuous control law, which reduces the system performance. In [15], parameter adaptive sliding mode control was proposed for the position tracking control problem of a Stewart platform to reduce the chattering phenomenon caused by traditional sliding mode control method. A new Lyapunov candidate functional containing an absolute function with fractional powers of the switching surface was proposed in [16], which makes the designed control law continuous and smooth, and eliminates the chattering problem.

In this article, the flexible joint space robot system is decomposed with singular perturbation method which becomes two subsystems: fast and slow. For the slow subsystem, a nonlinear integral sliding mode control is proposed, which designs a nonlinear integral sliding mode

This work is supported by National Natural Science Foundation of China (NSFC) under Grants 62403242 and 61973167, "the Fundamental Research Funds for the Central Universities", No. 30924010932.

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surface to avoid the integral saturation phenomenon generated by the introduction of integral and suppress chattering, and adopts the fast reaching law to design the sliding mode control law so as to ensure the tracking control performance. For the fast subsystem, speed difference feedback control is used to suppress the elastic vibration of flexible joints. The simulation is performed with two-link flexible joint space robot system, and it is verified that the method can improve the system control ability and robustness.

The article is structured as follows. Section 2 establishes the dynamic model of two-link flexible joint space robot with controlled base attitude. In Section 3, according to the singular perturbation theory, the system is split into slow and fast parts, and the speed difference feedback controller and nonlinear integral sliding mode controller are respectively designed, and demonstrate its stability. Section 4 verifies the effectiveness of the proposed method through simulation. Section 5 concludes the article in the end.

## 2 Dynamic modeling

The planar two-link flexible joint space robot system model is simplified as shown in Fig. 1, and its associated symbol representation is shown in Table 1.

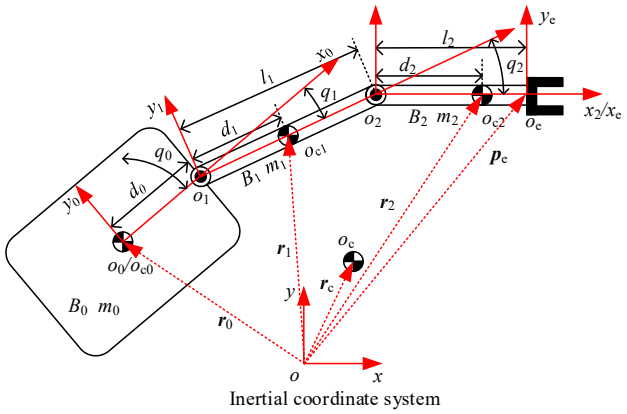



Fig. 1: System model of flexible joint space robot

Table 1: Symbol representation of flexible joint space robot

| Symbol | Meaning |
|---|---|
| $B_0$ | Floating base platform of the space robot |
| $B_1$ , $B_2$ | Two links |
| $m_0$ | Mass of the base |
| $m_1$ , $m_2$ | Masses of two links |
| $o_{ci}(i=0,1,2)$ | Position of each center of mass of the base and links |
| $o_c$ | Center of mass of the whole system |
| $o_1$ , $o_2$ | Flexible joints of space robot |
| $OXY$ | Inertial coordinate system of the translational motion |
| $O_iX_iY_i$ | Coordinate system of each component |
| | $B_i(i=0,1,2)$ |
| $O_eX_eY_e$ | Coordinate system of the end-effector |
| $l_1$ , $l_2$ | Length of two links |
| $d_i(i=0,1,2)$ | Distance from the mass center of each part to the joint |
| $\boldsymbol{r}_i(i=0,1,2)$ | Position vector of the center of mass in $OXY$ |
| $\boldsymbol{p}_e$ | Position vector of the end-effector in $OXY$ |
| $q_i(i=0,1,2)$ | Rotational angles of the base and links |

The flexible coupling between the manipulator arm and the drive motor rotor installed at the joint to drive its motion can be simplified as a linear spring with a stiffness coefficient of $K$ for robotic systems with flexible joints, according to the assumptions made in [17], its simplified model is shown in Fig. 2. When the motor rotor at the joint $o_i(i=1,2)$ rotates through a certain angle, there is a difference between the output angle $\theta_i(i=1,2)$ of the joint drive device and the actual angle $q_i(i=0,1,2)$ of the manipulator $B_i(i=1,2)$ driven by the joint drive device due to the elastic force of the spring, and the difference is $\theta_i - q_i$. Therefore, the effect of joint flexibility needs to be considered when designing the controller so that the stability of flexible joint space robot can be ensured.

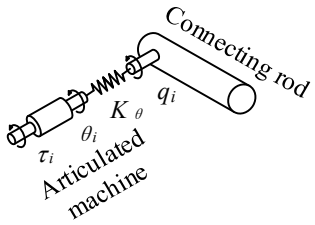



Fig. 2: Flexible Joint Model

Dynamic equation of flexible joint space robot with controllable base attitude is established as follows by using Lagrange method:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} = \begin{bmatrix} \tau_0 \\ \boldsymbol{\tau}_\theta \end{bmatrix} \tag{1}$$

$$\boldsymbol{J}\ddot{\boldsymbol{\theta}} + \boldsymbol{\tau}_\theta = \boldsymbol{\tau} + \boldsymbol{\tau}_d \tag{2}$$

$$\boldsymbol{\tau}_\theta = \boldsymbol{K}_\theta(\boldsymbol{\theta} - \boldsymbol{q}_\theta) \tag{3}$$

where $\boldsymbol{q} = [q_0 \quad \boldsymbol{q}_\theta]^{\mathrm{T}}$, $\boldsymbol{q}_\theta = [q_1 \quad q_2]^{\mathrm{T}}$, $\boldsymbol{\theta} = [\theta_1 \quad \theta_2]^{\mathrm{T}}$ are the rotation angles of the two joint drive motors, $\boldsymbol{J} = \mathrm{diag}(J_{\theta 1}, J_{\theta 2})$ is the diagonal and positive definite inertia matrix of the drive motor, $\boldsymbol{K}_\theta = \mathrm{diag}(K_{\theta 1}, K_{\theta 2})$ is the matrix of stiffness coefficients of the flexible joint after simplified as a linear spring, $\boldsymbol{M}(\boldsymbol{q}) \in R^{3\times 3}$ denotes the matrix of inertia of the flexible joint space robot, $\boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}}) \in R^{3\times 3}$ denotes the Coriolis force and centripetal force matrix, $\boldsymbol{\tau}$ means the output torque of the joint drive motor, $\boldsymbol{\tau}_\theta$ is the driving linkage torque generated by the joint flexibility, $\tau_0$ means the base attitude control torque,


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$\boldsymbol{\tau}_d$ denotes the disturbance torque such as motor side perturbation and nonlinear friction.

## 3 Controller Design

### 3.1 Fast subsystem controller design

Based on the system model established by the above dynamic relationship, the singular perturbation theory is used to approximately decompose the system dynamics equations (1)-(3) into mutually independent slow subsystem representing the rigid motion part and fast subsystem representing the flexible motion part, and respectively design appropriate control algorithms for the subsystems. Thus, the total control law of the system is expressed as:

$$\boldsymbol{\tau} = \boldsymbol{\tau}_s + \boldsymbol{\tau}_f \tag{4}$$

In equation (4), the slow subsystem control law $\boldsymbol{\tau}_s$ is designed for realizing the asymptotic tracking of the desired trajectory of the system, and the fast subsystem control law $\boldsymbol{\tau}_f$ is designed for suppressing the flexible vibration of the system caused by the flexible joints and ensure the stability.

Define the positive scale factor $\varepsilon$ and the positive definite diagonal matrix $\boldsymbol{K}_1$, which satisfy the following relationship with the stiffness coefficient matrix $\boldsymbol{K}_\theta$:

$$\boldsymbol{K}_\theta = \frac{\boldsymbol{K}_1}{\varepsilon^2} \tag{5}$$

Let the joint angle of the manipulator $\boldsymbol{q}_\theta$ be the slow variable, and the elastic force $\boldsymbol{z} = \boldsymbol{K}_\theta(\boldsymbol{\theta} - \boldsymbol{q}_\theta)$ of the flexible spring of the joint be the fast variable, and according to the above system dynamic equation, the fast subsystem equation which represents the elastic vibration of the system can be obtained:

$$\varepsilon^2 \ddot{\boldsymbol{z}} = \boldsymbol{J}^{-1}\boldsymbol{K}_1(\boldsymbol{\tau} + \boldsymbol{\tau}_d - \boldsymbol{J}\ddot{\boldsymbol{q}}_\theta - \boldsymbol{z}) \tag{6}$$

Design speed difference feedback control law:

$$\boldsymbol{\tau}_f = \boldsymbol{K}_f(\dot{\boldsymbol{q}}_\theta - \dot{\boldsymbol{\theta}}) \tag{7}$$

where $\boldsymbol{K}_f = \boldsymbol{K}_2 / \varepsilon$, $\boldsymbol{K}_2$ is a positive definite and diagonal parameter matrix.

Combining equations (4)(6)(7), the following dynamic equation for fast subsystem can be obtained:

$$\varepsilon^2 \boldsymbol{J}\ddot{\boldsymbol{z}} + \varepsilon \boldsymbol{K}_2 \dot{\boldsymbol{z}} + \boldsymbol{K}_1 \boldsymbol{z} = \boldsymbol{K}_1(\boldsymbol{\tau}_s - \boldsymbol{J}\ddot{\boldsymbol{q}}_\theta) \tag{8}$$

When $\varepsilon \to 0$, the stiffness coefficient matrix $\boldsymbol{K}_\theta \to \infty$, which means the flexible joint manipulator system can be equated to a rigid manipulator model, namely, $\boldsymbol{q}_\theta = \boldsymbol{\theta}$, $\boldsymbol{z} = 0$. Combining equations (1)(2)(3) the slow subsystem dynamic equation can be derived:

$$\boldsymbol{M}_\theta(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}_\theta(\boldsymbol{q})\dot{\boldsymbol{q}} = \boldsymbol{\tau}_s \tag{9}$$

where $\boldsymbol{M}_\theta(\boldsymbol{q}) = \boldsymbol{M}(\boldsymbol{q}) + \boldsymbol{J}_x$, $\boldsymbol{J}_x = \mathrm{diag}(0, J_{\theta 1}, J_{\theta 2})$, $\boldsymbol{C}_\theta(\boldsymbol{q})$ is the matrix resulting from $\dot{\boldsymbol{q}} = \dot{\boldsymbol{\theta}}$.

### 3.2 Slow subsystem controller design

Defining the tracking error vector of the manipulator joint angle as $\boldsymbol{e} = \boldsymbol{q} - \boldsymbol{q}_d$, in which the desired trajectory is $\boldsymbol{q}_d = [q_{d0}, q_{d1}, q_{d2}]^{\mathrm{T}}$, this article designs a nonlinear integral sliding mode surface as follows:

$$\boldsymbol{S} = \dot{\boldsymbol{e}} + \boldsymbol{A}\boldsymbol{e} + \boldsymbol{B}\int \boldsymbol{g}(\boldsymbol{e})dt \tag{10}$$

where $\boldsymbol{g}(\boldsymbol{e}) = [g(e_1), g(e_2), g(e_3)]^{\mathrm{T}}$, $g(e_i)$ $(i = 1,2,3)$ is a nonlinear function with the characteristic of amplifying the small error and saturating the large error, and its expression as follows:

$$g(e_i) = \begin{cases} \beta & e_i \geq \beta \\ \beta \sin\dfrac{\pi e}{2\alpha} & |e_i| < \beta \\ -\beta & e_i \leq -\beta \end{cases} \tag{11}$$

where the setup parameters are $\alpha \in R^+$, $\beta \in R^+$, and the derivation of the sliding mode surface can be obtained:

$$\dot{\boldsymbol{S}} = \ddot{\boldsymbol{e}} + \boldsymbol{A}\dot{\boldsymbol{e}} + \boldsymbol{B}\boldsymbol{g}(\boldsymbol{e}) \tag{12}$$

where $\boldsymbol{A}$ and $\boldsymbol{B}$ are adjustable parameters and both are positive definite diagonal matrices, $\boldsymbol{A} = \mathrm{diag}(a_1, a_2, a_3)$, $\boldsymbol{B} = \mathrm{diag}(b_1, b_2, b_3)$. The time for the system reaching steady state can be shortened and the steady state error can be reduced by adjusting $\boldsymbol{A}$ and $\boldsymbol{B}$ appropriately.

Through the slow dynamic equation (9) we can obtain:

$$\ddot{\boldsymbol{q}} = \boldsymbol{M}_\theta^{-1}(\boldsymbol{q})[\boldsymbol{\tau}_s - \boldsymbol{C}_\theta(\boldsymbol{q})\dot{\boldsymbol{q}}] \tag{13}$$

The equivalent controller based on equation (12) and equation (13) is designed as follows:

$$\boldsymbol{\tau}_{eq} = \boldsymbol{M}_\theta(\boldsymbol{q})[\ddot{\boldsymbol{q}}_d - \boldsymbol{A}\dot{\boldsymbol{e}} - \boldsymbol{B}\boldsymbol{g}(\boldsymbol{e})] + \boldsymbol{C}_\theta(\boldsymbol{q})\dot{\boldsymbol{q}} \tag{14}$$

To mitigate the high-frequency chattering of the sliding mode control, a switching controller based on the fast control law is designed in the following form:

$$\boldsymbol{\tau}_{sw} = -\boldsymbol{k}_0 \|\boldsymbol{S}\| \mathrm{sgn}(\boldsymbol{S}) - \boldsymbol{k}_1 S \tag{15}$$

where $\boldsymbol{k}_0$ and $\boldsymbol{k}_1$ are positive definite diagonal matrices. The isochronous reaching term $-\boldsymbol{k}_0 \|\boldsymbol{S}\| \mathrm{sgn}(\boldsymbol{S})$ can ensure that the time of reaching the sliding mode is finite, the introduction of the exponential reaching term $-\boldsymbol{k}_1 \boldsymbol{S}$ can shorten the reaching time. When the motion of the system state is far away from the sliding mode surface, $\|\boldsymbol{S}\|$ is larger, the moving point will converge to the sliding mode with faster speed. When the point is close to the sliding mode surface, $\|\boldsymbol{S}\|$ is smaller, and it enters into the sliding mode surface smoothly under the joint effect of isochronous reaching term and exponential reaching term, meanwhile, the chattering amplitude of the moving point will be attenuated, finally stabilized at the origin. Therefore, adjusting $\boldsymbol{k}_0$ and $\boldsymbol{k}_1$ reasonably can ensure the fast convergence while weakening the chattering.

Combining equations (14) and (15), the control law for slow subsystem can be given as:

$$\begin{aligned} \boldsymbol{\tau}_s = {} & \boldsymbol{M}_\theta(\boldsymbol{q})[\ddot{\boldsymbol{q}}_d - \boldsymbol{A}\dot{\boldsymbol{e}} - \boldsymbol{B}\boldsymbol{g}(\boldsymbol{e})] + \boldsymbol{C}_\theta(\boldsymbol{q})\dot{\boldsymbol{q}} \\ & -\boldsymbol{k}_0 \|\boldsymbol{S}\| \mathrm{sgn}(\boldsymbol{S}) - \boldsymbol{k}_1 \boldsymbol{S} \end{aligned} \tag{16}$$

### 3.3 Stability analysis

The Lyapunov function is selected as follows:

$$V = \frac{1}{2}\boldsymbol{S}^{\mathrm{T}}\boldsymbol{S} \tag{17}$$

The derivation of equation (17) can be obtained:

$$\begin{aligned} \dot{V} = {} & \boldsymbol{S}^{\mathrm{T}}\dot{\boldsymbol{S}} \\ = {} & \boldsymbol{S}^{\mathrm{T}}\{[\boldsymbol{M}_\theta^{-1}(\boldsymbol{q})(\boldsymbol{\tau}_s - \boldsymbol{C}_\theta(\boldsymbol{q})\dot{\boldsymbol{q}})] \\ & -\ddot{\boldsymbol{q}}_d \boldsymbol{A}\dot{\boldsymbol{e}} + \boldsymbol{B}\boldsymbol{g}(\boldsymbol{e})\} \end{aligned} \tag{18}$$


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Combined with equation (16), we can obtain:

$$\begin{aligned}\dot{V} &= \boldsymbol{S}^{\mathrm{T}}\dot{\boldsymbol{S}} \\ &= \boldsymbol{S}^{\mathrm{T}}[-\boldsymbol{M}_{\theta}^{-1}(\boldsymbol{q})(\boldsymbol{k}_0\|\boldsymbol{S}\|\mathrm{sgn}(\boldsymbol{S})+\boldsymbol{k}_1\boldsymbol{S})] \\ &= -\boldsymbol{M}_{\theta}^{-1}(\boldsymbol{q})(\boldsymbol{k}_0\|\boldsymbol{S}\|^2+\boldsymbol{k}_1\|\boldsymbol{S}\|^2) \\ &\le 0\end{aligned} \tag{19}$$

Therefore, the nonlinear integral sliding mode controller is known to be stable by combining the Lyapunov stability criterion.

## 4 Simulation study

With a view to verifying the feasibility and effectiveness of the designed controller, numerical simulations were performed, and the simulations were all performed with the two-link flexible joint space robot model that is shown in Fig. 1.

The relevant model parameters are as follows: $m_0 = 40\mathrm{kg}$, $m_1 = m_2 = 2\mathrm{kg}$, $d_0 = d_1 = d_2 = 1.5\mathrm{m}$, $l_1 = 3\mathrm{m}$, the inertial torques in the center of each part are: $J_0 = 35\mathrm{kg}\cdot\mathrm{m}^2$, $J_1 = 1.5\mathrm{kg}\cdot\mathrm{m}^2$, $J_2 = 0.75\mathrm{kg}\cdot\mathrm{m}^2$.

The rotational inertia of the flexible joint drive motor and its joint stiffness are:

$J_{\theta 1} = J_{\theta 2} = 0.07\mathrm{kg}\cdot\mathrm{m}^2$, $K_{\theta 1} = K_{\theta 2} = 500\mathrm{N}\cdot\mathrm{m}/\mathrm{rad}$.

The traditional sliding mode surface is chosen as $\boldsymbol{S} = \dot{\boldsymbol{e}} + \boldsymbol{A}_1\boldsymbol{e}$, and its control law is the same as the nonlinear integral sliding mode. The coefficients of the nonlinear sliding mode control law are set as:

$\boldsymbol{A} = \mathrm{diag}(4,4,4)$, $\boldsymbol{B} = \mathrm{diag}(150,150,150)$,

$\boldsymbol{k}_0 = \mathrm{diag}(2,2,2)$, $\boldsymbol{k}_1 = \mathrm{diag}(200,200,200)$.

The coefficients of the traditional sliding mode control law are set as:

$\boldsymbol{A}_1 = \mathrm{diag}(6,6,6)$,

$\boldsymbol{k}_0 = \mathrm{diag}(2,2,2)$, $\boldsymbol{k}_1 = \mathrm{diag}(200,200,200)$.

The initial angles of the base and manipulators are set: $\boldsymbol{q} = [0.03 \quad 0.06 \quad -0.12]^{\mathrm{T}}$,and the initial angles of the drive motors are set: $\boldsymbol{\theta} = [0.06 \quad -0.12]^{\mathrm{T}}$.

The motor-side disturbances are selected as follows as the mixed signal consisting of normal-valued disturbances and periodic disturbances with two frequencies:

$$\boldsymbol{\tau}_d = \begin{bmatrix} 4+3\sin(0.2\pi t)-2\cos(0.4\pi t) \\ -3+4\sin(0.2\pi t)-3\sin(0.4\pi t) \end{bmatrix}(\mathrm{N}\cdot\mathrm{m})$$

The desired trajectories are given respectively for the base attitude angle, the manipulator rotation angle 1 and the manipulator rotation angle 2:

$$\begin{cases} q_{d0} = 0.1\sin\left(\dfrac{\pi}{8}t\right) \\ q_{d1} = 0.1\sin\left(\dfrac{\pi}{8}t+\dfrac{2}{3}\pi\right)(rad) \\ q_{d2} = 0.1\sin\left(\dfrac{\pi}{8}t+\dfrac{4}{3}\pi\right) \end{cases}$$

The simulation results are shown in Figs. 3-6. Fig. 3 shows the trajectory tracking curves of the base attitude angle and the joint angles of the two manipulators in this article, and it can be observed that the actual rotation angles of each part can follow the desired trajectory in about 1.5s. Fig. 4 represents the tracking errors comparison graph between the nonlinear integral sliding mode control which is proposed in this article and the traditional sliding mode control in each part of the rotation angle, it can be shown that the error of the nonlinear integral sliding mode control method converges faster and the steady-state error is smaller than traditional sliding mode control. Additionally, the traditional sliding mode error has chattering during convergence, whereas the nonlinear integral sliding mode has weakened the chattering phenomenon. Fig. 5 represents the torques that drive the rotation of the three parts which are the base and the manipulators, and it can be found that the input torque of each part is steady, smooth and basically without chattering. Fig. 6 indicates the angular plot of the driving motor rotor, and it can be similarly shown that the tracking angle tends to a normalized state in about 1.5s and maintains smooth and stable tracking.

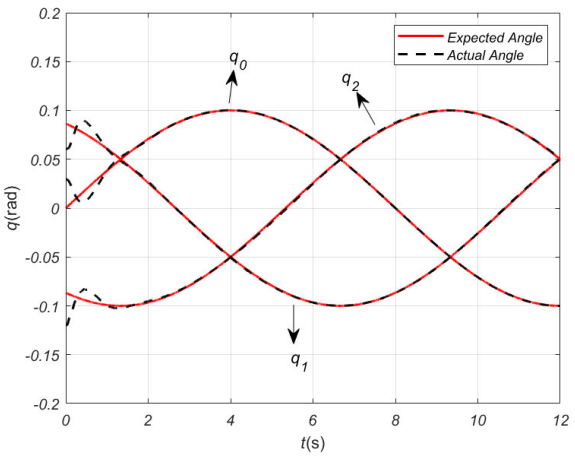



Fig. 3: Joint angle trajectory tracking curve

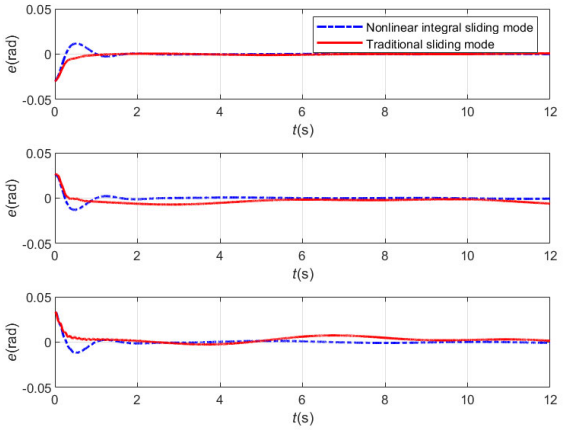



Fig. 4: Comparison of joint angle errors


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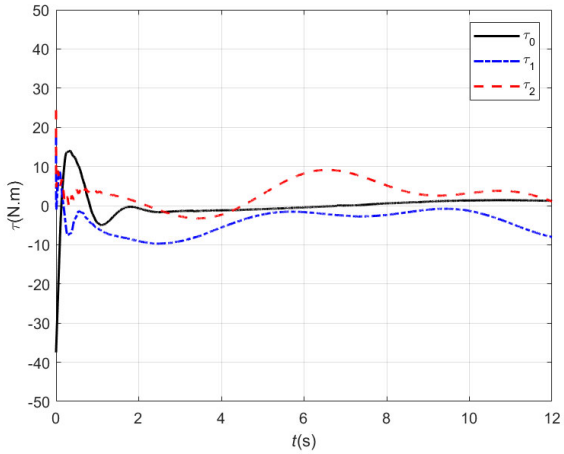



Fig. 5: Joint driving torque diagram

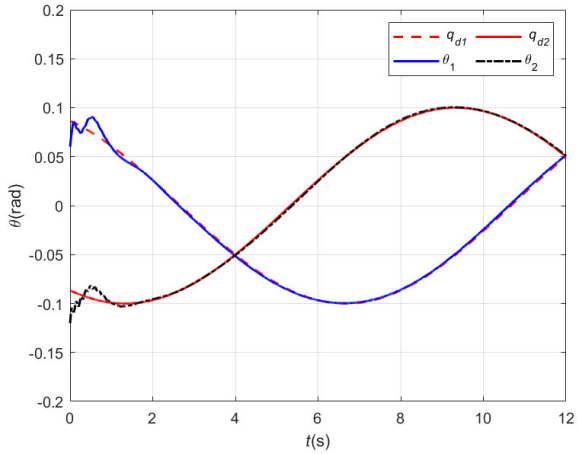



Fig. 6: Rotor angle diagram of drive motor

## 5 Conclusion

In this article, a nonlinear integral sliding mode control optimization method is proposed for the tracking control problem of flexible joint space robot with controlled base attitude under external disturbances. The dynamic model of the flexible joint space robot is established based on the Lagrange method, and the system model is decomposed into two subsystems, fast and slow, through the singular perturbation theory. For the slow subsystem, a nonlinear function is introduced to design the integral sliding mode controller and its stability is demonstrated. The designed speed difference feedback control algorithm is applied to the fast subsystem. The simulation results indicate that the proposed method has a satisfactory tracking performance, reduces the chattering and enhances the robustness of the system.

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